

Fundamental Partial Differential Equations

Solving the 2D Incompressible Navier-Stokes Equations Lid-Driven Cavity Flow

Abstract

This report introduces the numerical solution of the two-dimensional incompressible Navier–Stokes equations. Building on previous work on convection, diffusion, Burgers' equation, and the Poisson equation, the report shows how these components combine to form a complete incompressible flow solver. The lid-driven cavity problem is used as the canonical example because it is geometrically simple but captures the essential coupling between nonlinear convection, viscous diffusion, pressure, and incompressibility.

1. Introduction

The previous topics developed the main mathematical ingredients required for solving incompressible fluid flow:

convection $\rightarrow$ diffusion $\rightarrow$ Burgers' equation $\rightarrow$ Poisson equation.

The incompressible Navier–Stokes equations combine these ideas into a coupled system for velocity and pressure.

The main new feature is that the velocity field must satisfy the incompressibility condition:

$$\nabla \cdot \mathbf{u} = 0.$$

This condition is enforced through the pressure field, which is obtained by solving a pressure Poisson equation.

2. Governing Equations

Let the two-dimensional velocity field be

$$\mathbf{u}(x, y, t) = \begin{bmatrix} u(x, y, t) \\ v(x, y, t) \end{bmatrix},$$

where:

- u is the velocity component in the x -direction,
- v is the velocity component in the y -direction.

The two-dimensional incompressible Navier–Stokes equations are

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \\ \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right). \end{aligned}$$

The incompressibility condition is

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

In vector notation, the momentum equation is

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}.$$

Together with

$$\nabla \cdot \mathbf{u} = 0,$$

this forms the incompressible Navier–Stokes system.

3. Physical Interpretation of Each Term

The momentum equation can be written as

$$\underbrace{\frac{\partial \mathbf{u}}{\partial t}}_{\text{unsteady term}} + \underbrace{(\mathbf{u} \cdot \nabla) \mathbf{u}}_{\text{nonlinear convection}} = \underbrace{-\frac{1}{\rho} \nabla p}_{\text{pressure force}} + \underbrace{\nu \nabla^2 \mathbf{u}}_{\text{viscous diffusion}}.$$

Each term has a distinct physical role:

- The unsteady term describes temporal change in the velocity field.
- The nonlinear convection term describes transport of momentum by the flow itself.
- The pressure-gradient term redistributes momentum and enforces incompressibility.
- The viscous diffusion term smooths velocity gradients.

The incompressibility condition

$$\nabla \cdot \mathbf{u} = 0$$

ensures that fluid volume is locally conserved.

4. Lid-Driven Cavity Flow

The lid-driven cavity is a standard benchmark problem for incompressible flow.

The domain is a square cavity:

$$\Omega = [0, L] \times [0, L].$$

The top wall moves with a prescribed horizontal velocity, while the other walls remain stationary.

This simple configuration generates a circulating flow inside the cavity.

5. Boundary Conditions

For the lid-driven cavity problem, the velocity boundary conditions are:

Top wall

$$u = U, \quad v = 0.$$

Bottom wall

$$u = 0, \quad v = 0.$$

Left wall

$$u = 0, \quad v = 0.$$

Right wall

$$u = 0, \quad v = 0.$$

These are no-slip boundary conditions. The top wall drives the flow by imposing a non-zero tangential velocity.

Pressure boundary conditions are also required. A common choice is:

$$\begin{aligned} \frac{\partial p}{\partial x} &= 0 && \text{on the left and right walls,} \\ \frac{\partial p}{\partial y} &= 0 && \text{on the bottom wall,} \end{aligned}$$

and

$$p = 0 \quad \text{on the top wall.}$$

The pressure reference value is needed because pressure in incompressible flow is determined only up to an arbitrary constant.

6. Discretisation of the Momentum Equations

Let the computational grid be

$$x_i = i\Delta x, \quad y_j = j\Delta y,$$

with

$$i = 0, 1, \dots, N_x - 1, \quad j = 0, 1, \dots, N_y - 1.$$

The velocity and pressure are approximated as

$$u(x_i, y_j, t^n) \approx u_{i,j}^n,$$

$$v(x_i, y_j, t^n) \approx v_{i,j}^n,$$

$$p(x_i, y_j, t^n) \approx p_{i,j}^n.$$

The convective terms are approximated using backward differences:

$$\frac{\partial u}{\partial x} \approx \frac{u_{i,j}^n - u_{i-1,j}^n}{\Delta x},$$

$$\frac{\partial u}{\partial y} \approx \frac{u_{i,j}^n - u_{i,j-1}^n}{\Delta y}.$$

The diffusion terms are approximated using central differences:

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{\Delta x^2},$$

$$\frac{\partial^2 u}{\partial y^2} \approx \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{\Delta y^2}.$$

Analogous expressions are used for the v -velocity component.

7. Pressure Poisson Equation

The pressure field is obtained from a Poisson equation:

$$\nabla^2 p = b.$$

The source term b is constructed from the velocity field. A commonly used form is

$$b = \rho \left[\frac{1}{\Delta t} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) - \left(\frac{\partial u}{\partial x} \right)^2 - 2 \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} - \left(\frac{\partial v}{\partial y} \right)^2 \right].$$

The pressure Poisson equation in two dimensions is

$$\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} = b.$$

Using central differences gives

$$\frac{p_{i+1,j} - 2p_{i,j} + p_{i-1,j}}{\Delta x^2} + \frac{p_{i,j+1} - 2p_{i,j} + p_{i,j-1}}{\Delta y^2} = b_{i,j}.$$

Solving for $p_{i,j}$ gives

$$p_{i,j} = \frac{\Delta y^2 (p_{i+1,j} + p_{i-1,j}) + \Delta x^2 (p_{i,j+1} + p_{i,j-1}) - b_{i,j} \Delta x^2 \Delta y^2}{2(\Delta x^2 + \Delta y^2)}.$$

The pressure equation is solved iteratively at every time step.

8. Velocity Update Equations

Once the pressure field has been computed, the velocity components are updated.

The update for u is

$$\begin{aligned} u_{i,j}^{n+1} = & u_{i,j}^n - u_{i,j}^n \frac{\Delta t}{\Delta x} (u_{i,j}^n - u_{i-1,j}^n) - v_{i,j}^n \frac{\Delta t}{\Delta y} (u_{i,j}^n - u_{i,j-1}^n) \\ & - \frac{\Delta t}{2\rho\Delta x} (p_{i+1,j}^n - p_{i-1,j}^n) \\ & + \nu \frac{\Delta t}{\Delta x^2} (u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n) \\ & + \nu \frac{\Delta t}{\Delta y^2} (u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n). \end{aligned}$$

The update for v is

$$\begin{aligned} v_{i,j}^{n+1} = & v_{i,j}^n - u_{i,j}^n \frac{\Delta t}{\Delta x} (v_{i,j}^n - v_{i-1,j}^n) - v_{i,j}^n \frac{\Delta t}{\Delta y} (v_{i,j}^n - v_{i,j-1}^n) \\ & - \frac{\Delta t}{2\rho\Delta y} (p_{i,j+1}^n - p_{i,j-1}^n) \\ & + \nu \frac{\Delta t}{\Delta x^2} (v_{i+1,j}^n - 2v_{i,j}^n + v_{i-1,j}^n) \\ & + \nu \frac{\Delta t}{\Delta y^2} (v_{i,j+1}^n - 2v_{i,j}^n + v_{i,j-1}^n). \end{aligned}$$

These updates combine nonlinear convection, pressure forcing, and viscous diffusion.

9. Numerical Algorithm

The full numerical procedure is:

1. Define the computational domain.
2. Choose N_x , N_y , Δx , Δy , and Δt .
3. Initialise u , v , and p .
4. Apply velocity boundary conditions.
5. For each time step:
 - copy the current velocity fields,
 - build the pressure source term b ,
 - solve the pressure Poisson equation,
 - update the u -velocity,
 - update the v -velocity,
 - reapply velocity boundary conditions,
 - reapply pressure boundary conditions.
6. Continue until the final time or steady state is reached.

Symbolically, the time-stepping loop is

$$\text{build } b \rightarrow \text{solve } \nabla^2 p = b \rightarrow \text{update } u, v \rightarrow \text{apply boundary conditions.}$$

10. Stability Considerations

The time step must satisfy stability restrictions.

For convection, the CFL condition is

$$\max |u| \frac{\Delta t}{\Delta x} + \max |v| \frac{\Delta t}{\Delta y} \leq 1.$$

For diffusion, a typical explicit stability condition is

$$\nu \Delta t \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} \right) \leq \frac{1}{2}.$$

The time step must be chosen to satisfy both constraints.

If the time step is too large, the numerical solution may become unstable.

11. Expected Flow Behaviour

At early times, the moving lid begins to drag the fluid near the top boundary.

As time progresses:

- momentum diffuses into the cavity,
- the fluid begins to circulate,
- a primary vortex forms near the centre of the cavity,
- smaller secondary vortices may appear near the corners at sufficiently high Reynolds number.

The Reynolds number is

$$Re = \frac{UL}{\nu}.$$

For low Reynolds number, viscous effects dominate and the flow is smooth.

For higher Reynolds number, inertial effects become stronger and the vortex structure becomes more pronounced.

12. Connection to Previous Topics

The Navier--Stokes solver combines all previous topics.

From convection

The nonlinear transport term is

$$(\mathbf{u} \cdot \nabla) \mathbf{u}.$$

From diffusion

The viscous smoothing term is

$$\nu \nabla^2 \mathbf{u}.$$

From Burgers' equation

The combination of nonlinear convection and diffusion is

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = \nu \nabla^2 \mathbf{u}.$$

From the Poisson equation

The pressure field is obtained from

$$\nabla^2 p = b.$$

New in Navier--Stokes

The pressure gradient and incompressibility constraint are added:

$$-\frac{1}{\rho} \nabla p, \quad \nabla \cdot \mathbf{u} = 0.$$

Thus, the incompressible Navier--Stokes equations may be viewed as Burgers' equations plus pressure and incompressibility.

13. Validation and Diagnostics

A numerical solution should be assessed using several checks.

Divergence check

The computed velocity field should satisfy

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \approx 0.$$

This verifies incompressibility.

Boundary condition check

The velocity values on the boundaries should match the prescribed wall velocities.

Steady-state check

For steady cavity flow, the velocity field should eventually stop changing significantly:

$$\|u^{n+1} - u^n\| < \varepsilon, \quad \|v^{n+1} - v^n\| < \varepsilon.$$

Physical behaviour check

The solution should show recirculating motion inside the cavity, with the strongest horizontal velocity near the moving lid.

14. Summary

This report described the mathematical and numerical formulation of a two-dimensional incompressible Navier--Stokes solver for lid-driven cavity flow.

The main ideas are:

- the velocity field has two components, u and v ,
- nonlinear convection transports momentum,
- viscosity diffuses momentum,
- pressure enforces incompressibility,
- the pressure field is obtained from a Poisson equation,
- boundary conditions define the physical problem,
- the lid-driven cavity produces a recirculating flow.

The numerical solution follows the repeated sequence

build pressure source $\rightarrow$ solve pressure Poisson equation $\rightarrow$ update velocity $\rightarrow$ apply boundary conditions.

This completes the transition from model PDEs to a full incompressible flow solver.

15. Roadmap Position

The learning pathway is now

1D convection $\rightarrow$ 1D diffusion $\rightarrow$ 1D Burgers' $\rightarrow$ 2D convection $\rightarrow$ 2D diffusion $\rightarrow$ 2D Burgers' $\rightarrow$ Laplace equation $\rightarrow$ Poisson ec

The next natural extension is to implement the solver computationally and visualise:

- velocity magnitude,
- pressure contours,
- streamlines,
- vorticity,
- divergence error.